

- 3 (a) In one second, the volume of air that is swept by the blades is

$$\begin{aligned} V &= A \times v \quad (1) \\ &= \pi (50)^2 \times 20 \\ &= 1.57 \times 10^5 \text{ m}^3 \end{aligned}$$

- (b) In 1 s, mass = ρV

$$\begin{aligned} &= 1.20 \times 1.57 \times 10^5 \\ &= 1.88 \times 10^5 \text{ kg} \end{aligned}$$

(c) (i) Rate of loss KE = $\frac{1}{2} \left(\frac{dm}{dt} \right) (u^2 - v^2)$

$$\begin{aligned} &= \frac{1}{2} \times 1.88 \times 10^5 \times (20^2 - 15^2) \\ &= 1.65 \times 10^7 \text{ W} \end{aligned}$$

(ii) Force = $\frac{dm}{dt} \times (u - v)$

$$\begin{aligned} &= 1.88 \times 10^5 \times (20 - 15) \\ &= 9.42 \times 10^5 \text{ N} \end{aligned}$$

- (d) The rate of loss of KE is equal to the product of the force and average wind speed.

- 4 (a) (i) By the principle of conservation